

JEE ADVANCED CUMULATIVE TEST SERIES

44
YEARS
OF EXCELLENCE



NARAYANA
IIT-JEE ACADEMY-INDIA

Sec: JR.IIT_*CO-SC(MODEL-A)

Time: 3 Hrs

Date: 23-06-24

Max. Marks: 186

23-06-24_JR.IIT_STAR CO-SC(MODEL-A)_JEE ADV_CAT-6_SYLLABUS

PHYSICS:

(100% CUMULATIVE)

SYLLABUS COVERED FROM 15-04-24 TO 22-06-24

CHEMISTRY:

(100% CUMULATIVE)

SYLLABUS COVERED FROM 15-04-24 TO 22-06-24

MATHEMATICS: (100% CUMULATIVE)

SYLLABUS COVERED FROM 15-04-24 TO 22-06-24

THE NARAYANA GROUP

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 6)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 7 – 14)	Questions with Multiple Correct Choice (Partial Marking +1)	4	-2	8	32
Sec – III(Q.N : 15 – 18)	Questions with Comprehension Type (2 Comprehension-2+2=4Q)	3	0	4	12
Total				18	62

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 24)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 25 – 32)	Questions with Multiple Correct Choice (Partial Marking +1)	4	-2	8	32
Sec – III(Q.N : 33 – 36)	Questions with Comprehension Type (2 Comprehension-2+2=4Q)	3	0	4	12
Total				18	62

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 42)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 43 – 50)	Questions with Multiple Correct Choice (Partial Marking +1)	4	-2	8	32
Sec – III(Q.N : 51 – 54)	Questions with Comprehension Type (2 Comprehension-2+2=4Q)	3	0	4	12
Total				18	62

SECTION-I

(Single Correct Answer Type)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

- Let α and β be the roots of the equation $x^2 - x - 1 = 0$. If $P_n = 2024\alpha^n + 2026\beta^n$, $n \geq 1$, then the value of $P_3 - 2P_1$ is not divisible by
 A) 10 B) 35 C) 27 D) 45
- The value of $2\sin\left(\frac{\pi}{8}\right)\sin\left(\frac{2\pi}{8}\right)\sin\left(\frac{3\pi}{8}\right)\sin\left(\frac{5\pi}{8}\right)\sin\left(\frac{6\pi}{8}\right)\sin\left(\frac{7\pi}{8}\right)$ is
 A) $\frac{1}{4\sqrt{2}}$ B) $\frac{1}{4}$ C) $\frac{1}{8}$ D) $\frac{1}{8\sqrt{2}}$
- Let $\alpha \in \left(0, \frac{\pi}{2}\right)$ and $x^2 - (\sin \alpha)x + \cos^2 \alpha = 0$ has real roots and if $2\sin^2 \alpha - 9\sin \alpha \cos \alpha + 9\cos^2 \alpha = 0$ then the value of $\frac{(3+\sqrt{3})\sin \alpha + (3-\sqrt{3})\cos \alpha}{(3+\sqrt{3})\cos \alpha - (3-\sqrt{3})\sin \alpha}$ is equal to
 A) $\frac{5-\sqrt{3}}{-5-3\sqrt{3}}$ B) $\frac{5-\sqrt{3}}{-5+3\sqrt{3}}$ C) $\frac{5\sqrt{3}}{3\sqrt{3}-5}$ D) $\frac{5+\sqrt{3}}{5-\sqrt{3}}$
- $1^2 - 2^2 + 3^2 - 4^2 + 5^2 - 6^2 + \dots$ upto 2027 terms = 2027 m, where m is an integer. Then number of common terms of the two A.P's 3, 6, 9, 15, m and 5, 9, 13, 4005 is
 A) 84 B) 74 C) 94 D) 64
- Let $a_1, a_2, \dots, a_n$ be a given A.P. whose common difference is an integer and $S_n = a_1 + a_2 + \dots + a_n$. If $a_1 = 1$, $a_n = 300$ and $15 \leq n \leq 50$, then the ordered pair (S_{n-4}, a_{n-4}) is equal to :
 A) (2480, 249) B) (2490, 249) C) (2490, 248) D) (2480, 248)
- The set of all real values of λ for which the quadratic equations, $(\lambda^2 + 1)x^2 - 4\lambda x + 2 = 0$ always have exactly one root in the interval (0,1) is :
 A) (-3, -1) B) (1, 3] C) (0, 2) D) (2, 4]

SECTION-II

(Multiple Correct Answer Type)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases.

7. 'N' represents set of all Natural numbers, $A = \{\alpha \in \mathbb{N} : \alpha^2 - 241\alpha + 2756 \leq 8\}$
- $B = \{\beta \in \mathbb{N} : 9e^{2x} - 6e^x + \beta = 0 \text{ has atleast one real solution for } x.\}$
- $C = (A - B) \cup (B - A)$, Based on the above information which of the following options are correct?
- A) These exists at-least one element in A which is not in B.
- B) There exist at-least one ordered pair (α, β) where $\alpha \in A$ and $\beta \in B$ such that $\alpha^2 + \beta^2 = 145$
- C) B is an infinite set
- D) $A \cap C = \phi$
8. If $A + B + C = \pi$, then which of the following options are correct?
- A) $\frac{\sin 2A + \sin 2B + \sin 2C}{4(\sin A + \sin B + \sin C)} = 2 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$ if $\sin A + \sin B + \sin C \neq 0$
- B) $\cot A \cot B + \cot B \cot C + \cot A \cot C = 1$ if A,B,C are not integral multiples of π .
- C) $\sin^2 A + \sin^2 B + \sin^2 C = 2 + 2 \cos A \cos B \cos C$
- D) $\tan A + \tan B + \tan C = \tan A \tan B \tan C$ if A,B,C are not odd multiples of $\frac{\pi}{2}$.
9. If A,B,C,D,E,F.....Z are 26 arithmetic means inserted between the numbers 1 and 325, $A < B < C < D < \dots X < Y < Z$ then which of the following statements are correct?
- [NOTE: ABCD means the product of the values of A,B,C,D].
- A) The median of the set of observations $\{A, N, A, G, H, A, A, N, V, I\}$ is 90
- B) $M + O + N + I + T + H$ is equal to 953
- C) The mean of the numbers $\{D, H, O, N, I, W, S, L, M, R\}$ is 163
- D) The value of CD is 1813

10. Let $f(\theta) = \left(1 + \frac{4\sin\theta}{\sin 6\theta}\right) \left(1 + \frac{4\sin 2\theta}{\sin 5\theta}\right)$, then:

- A) $f\left(\frac{\pi}{7}\right) = 25$ B) $f\left(\frac{\pi}{7}\right) = -25$ C) $f\left(\frac{2\pi}{7}\right) = 9$ D) $f\left(\frac{2\pi}{7}\right) = -9$

11. $X = \left\{x \in \mathbb{N} : \frac{((\sin x) - 10)(x^2 + 6x - 16)(x^2 - x + 1)}{x^2 - 64} \geq 0\right\}$

$Y = \left\{y \in \mathbb{N} : \frac{((\sin y) - 5)(3 - \cos y)(y^2 - 2y + 3)}{((\log_2 y) + 1)(y^2 - 16x + 48)} \leq 0\right\}$ 'N' is set of natural numbers

A) If S is the sum of all the elements in set X then the value of $\left(\sin \frac{9\pi}{S}\right)^2$ is $\frac{3}{4}$

B) $n(x) = 6$

C) $n(y) = 7$

D) Let K be the sum of all the elements in $(X \cap Y)$ then the value of

$\sin K^0 \cdot \sin(60 - K)^0 \cdot \sin(60 + K)^0$ is $\frac{1}{\sqrt{5} + 1}$

12. If A and B are acute angles such that $\tan A = \frac{1}{2}$ and $\tan B = \frac{1}{3}$, then:

A) $\sin^2(A + B) = \frac{1}{2}$

B) $\tan\left(\frac{A+B}{2}\right) = \sqrt{2} - 1$

C) $\cot\left(\frac{A+B}{2}\right) = 2 - \sqrt{3}$

D) $\cos(2A + 2B) = 0$

13. If the greatest value of $f(x) = -x^2 + 4x + \lambda - 4$, where $x \in [0, 5]$ is smaller than the least value of $g(x) = x^2 - 2\lambda x + 10 - 2\lambda$, where $x \in \mathbb{R}$ then λ may be:

A) $\frac{-3}{2}$

B) $\frac{-17}{4}$

C) $\frac{3}{11}$

D) $\frac{-1}{8}$

14. Let $x = \sin\theta \cos^3\theta$ and $y = \sin^3\theta \cos\theta$, then:

A) If $0 < \theta < \frac{\pi}{4}$, then $x + y > 0$

B) If $\frac{\pi}{4} < \theta < \frac{\pi}{2}$, then $x < y$

C) If $\frac{\pi}{2} < \theta < \frac{3\pi}{4}$, then $x + y > 0$

D) If $\frac{3\pi}{4} < \theta < \pi$, then $x < y$

SECTION-III (Paragraph Type)

This section contains 2 groups of question. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choice (A), (B), (C) AND (D) for its answer, out of which ONLY ONE is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

Paragraph for Questions 15 and 16:

Let V_r represents r^{th} term of a series, then

$$\sum_{r=1}^n V_r - V_{r-1} = V_1 - V_0 + V_2 - V_1 + V_3 - V_2 \dots V_n - V_{n-1} = V_n - V_0$$

15. Let $f(n) = \sum_{r=1}^n \log_{10} \left(\frac{9r+1}{9r-8} \right)$, then the value of $f(11)$ is

- A) 0 B) 2 C) -2 D) 4

16. Let $f_n(\theta) = \frac{\sin \theta}{\sin 2\theta \sin 3\theta} + \frac{\sin \theta}{\sin 3\theta \sin 4\theta} + \frac{\sin \theta}{\sin 4\theta \sin 5\theta} \dots n \text{ terms}$. Then the value of $f_{28}\left(\frac{\pi}{4}\right)$ is

- A) 0 B) 1 C) 2 D) not defined

Paragraph for Questions 17 and 18:

A relation R in a set A is called

- (i) reflexive, if $(a, a) \in R$, for every $a \in A$,
- (ii) symmetric, if $(a_1, a_2) \in R$ implies that $(a_2, a_1) \in R$, for all $a_1, a_2 \in A$.
- (iii) transitive, if $(a_1, a_2) \in R$ and $(a_2, a_3) \in R$ implies that $(a_1, a_3) \in R$, for all $a_1, a_2, a_3 \in A$.

Equivalence Relation: (A Relation is said to be equivalence if it is reflexive, symmetric, and transitive)

17. Let R be a relation defined on $\mathbb{N}$ as $a R b$ is $2a + 3b$ is a multiple of 5, $a, b \in \mathbb{N}$. Then R is

- A) not reflexive B) transitive but not symmetric
C) symmetric but not transitive D) An equivalence relation

18. Let R be a relation on $\mathbb{R}$, given by $R = \{(a, b) : 3a - 3b + \sqrt{7} \text{ is an irrational number}\}$. Then R is

- A) Reflexive but neither symmetric nor transitive
B) Reflexive and transitive but not symmetric
C) Reflexive and symmetric but not transitive
D) An equivalence relation

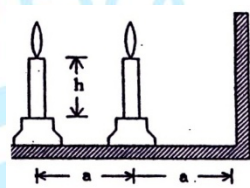
SECTION-I

(Single Correct Answer Type)

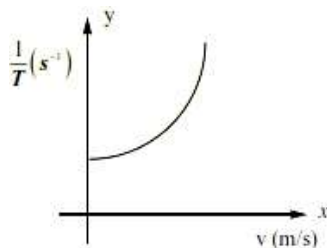
This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

19. Two candles of equal height 'h' at the initial moment are at distance of separation 'a'. The distance of nearest candle from the wall is also 'a' as shown. If the left candle burns down during time ' t_1 ' and the right candle during the time ' t_2 ', with what speed the shadow of tip of right candle move along the wall?



- A) $\left| \frac{h}{t_1} - \frac{2h}{t_2} \right|$ B) $\left| \frac{2h}{t_1} - \frac{2h}{t_2} \right|$ C) $\left| \frac{2h}{t_1} - \frac{h}{t_2} \right|$ D) $\left| \frac{h}{t_1} - \frac{h}{t_2} \right|$
20. A projectile is projected with speed u at an angle of 60° with horizontal from the foot of an inclined plane. If the projectile hits the inclined plane while moving horizontally, then range on inclined plane will be
- A) $\frac{u^2 \sqrt{21}}{2g}$ B) $\frac{3u^2}{4g}$ C) $\frac{u^2}{2\beta}$ D) $\frac{\sqrt{21}u^2}{8g}$
21. An engineer designs a robot that can climb stairs. Time T in which battery used in the robot is fully discharged depends on its constant speed v relative to the stairs. This dependence is shown in the graph which follows $y - 0.04 = x^2$. Determine the maximum length of the staircase that the robot can climb with constant velocity.

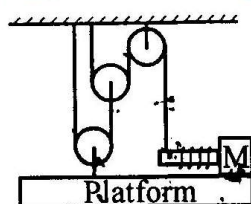


- A) 2.5 m B) 5m C) 7.5 D) 1.25 m

22. A train is running with uniform velocity in east direction. A car is running on a road parallel to the track with uniform speed. After some time, the road becomes perpendicular to railway track. The car driver notices that initial speed of train with respect to itself was 7 m/s and later on it became 13 m/s. What can be the true speed of the driver?

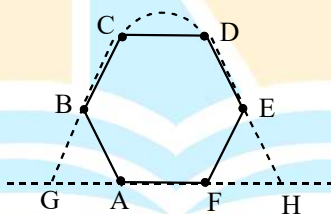
- A) 12 m/s B) 5 m/s C) 13 m/s D) 7 m/s

23. In the figure shown the velocity of platform while string is winding on the motor shaft with velocity 5m/s will be:



- A) 1 m/s B) 2 m/s C) 5 m/s D) 5/2 m/s

24. A particle is projected from the point G on the horizontal ground in the vertical plane such that it grazes the points B, C, D and E of a regular hexagon of side 'a'. Find its horizontal range GH.



- A) $3a$ B) $\sqrt{8}a$ C) $5a$ D) $\sqrt{7}a$

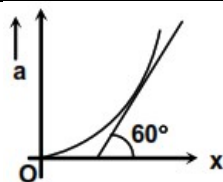
SECTION-II

(Multiple Correct Answer Type)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases.

25. A particle starts moving with initial velocity 3 m/s along x-axis from origin. Its acceleration is varying with x in parabolic nature ($a = kx^2$) as shown in figure. At $x = \sqrt{3}$ m tangent to the graph makes an angle 60° with positive x-axis as shown in diagram. Then at $x = \sqrt{3}$



A) $v = \sqrt{(\sqrt{3} + 9)} \text{ m/s}$

B) $a = 1.5 \text{ m/s}^2$

C) $v = \sqrt{12} \text{ m/s}$

D) $a = 3 \text{ m/s}^2$

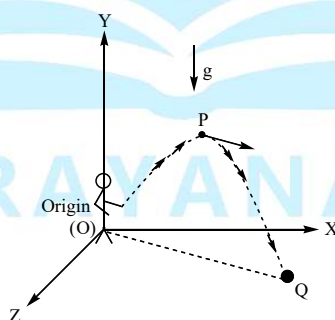
26. A train is moving at a speed of $v = 30 \text{ m/s}$ towards a stationary railway man next to the rails. The train starts whistling when it is at a distance $d = 360 \text{ m}$ from the man and does so for $T = 3 \text{ sec}$. The speed of sound in air $C = 300 \text{ m/s}$ the time duration for which the man hears the whistle is

- a) 2.4 sec b) 3.2 sec c) 2.7 sec d) independent of 'd'

27. A shell is fired from the origin with velocity $\vec{V} = 10\hat{i} + 10\hat{j} + 10\hat{k}$ in uniform gravity ($g = 10 \text{ m/s}^2$) acting downwards. In the diagram, P is the highest point and Q is the final point on the XZ plane. $\vec{V}_P$, $\vec{V}_Q$, $\vec{r}_P$ and $\vec{r}_Q$ represent velocity and position vector of point P and Q respectively.

$t_{OP} \rightarrow$ Time taken in going from O to P

$t_{PQ} \rightarrow$ Time taken in going from P to Q



Choose the correct option(s)

A) $\vec{V}_P \cdot \vec{V}_Q = 200$

B) $t_{OP} / t_{PQ} = 1$

C) $\vec{r}_P = 10\hat{i} + 5\hat{j} + 10\hat{k}$

D) $\vec{r}_Q = 20\hat{i} + 20\hat{k}$

28. A particle slides down a smooth inclined plane of elevation α . The incline is fixed end to end in an elevator of base length λ accelerating up with acceleration a_0 . Assume at $t = 0$ the particle is at the top of the incline then—

A) the particle has to travel a length $\lambda \cos \alpha$ with acceleration $(g + a_0) \sin \alpha$ down the incline in a time $\sqrt{\frac{\ell}{(g + a_0) \sin 2\alpha}}$

B) The particle has to travel a length $\frac{\ell}{\cos \alpha}$ with acceleration $g \sin \alpha$ down the incline in a time $\sqrt{\frac{2\ell}{a_0 \sin 2\alpha}}$

C) The particle has to travel a length $\frac{\ell}{\cos \alpha}$ with acceleration $(g + a_0) \sin \alpha$ down the incline in a time $\sqrt{\frac{4\ell}{(g + a_0) \sin(2\alpha)}}$

D) The incline offers a normal reaction $m(a_0 + g) \cos \alpha$ to the block so that it remains in contact with the incline.

29. Two masses of 10 kg and 20 kg are connected by a light spring as shown. A force of 200 N acts on a 20 kg mass as shown. At a certain instant the acceleration of 10 kg mass is 12 m/s^2 .



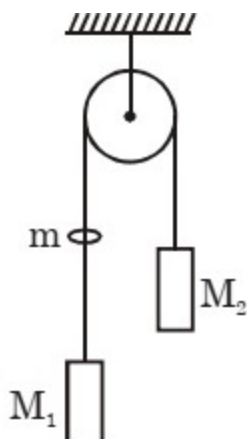
A) At that instant the 20 kg mass has an acceleration of 12 m/s^2

B) At that instant the 20 kg mass has an acceleration of 4 m/s^2

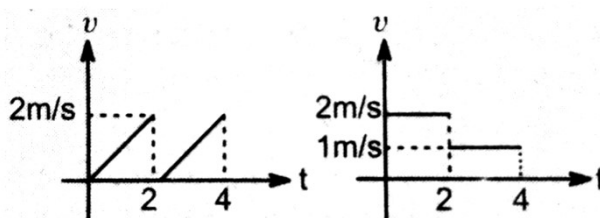
C) The stretching force in the spring is 120 N

D) The collective system moves with a common acceleration of 30 m/s^2 when the extension in the connecting spring is the maximum

30. In figure shown, the pulley and the string are massless. A bead of mass m can not slide on the string. (Given : $M_1 = 2\text{kg}, M_2 = 2\text{kg}, m = 1.5\text{kg}$)

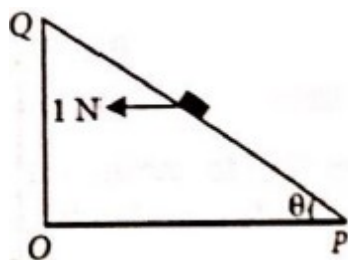


- A) The acceleration of the bead is downwards.
- B) The acceleration of M_1 is $\frac{3g}{11}$ downwards.
- C) The tension force on M_1 is $\frac{16g}{11}$
- D) The acceleration of M_2 is $\frac{g}{7}$ upwards
31. The velocity versus time graph of two particles moving along x-axis varies as shown in the following two plots. Then



- A) Maximum separation between the two particles is 2 m
- B) Maximum separation between the two particles is 2.5 m
- C) Maximum separation between occurs after time $t = 2\text{sec}$
- D) Maximum separation between occurs after time $t = 3\text{sec}$

32. A small block of mass of 0.1 kg lies on a fixed inclined plane PQ which makes an angle θ with the horizontal. A horizontal force of 1 N acts on the block through its center of mass as shown in the figure. The block remains stationary if (take $g = 10 \text{ m/s}^2$)



- A) $\theta = 45^\circ$
 B) $\theta > 45^\circ$ and a friction force acts on the block towards P.
 C) $\theta > 45^\circ$ and a frictional force acts on the block towards Q.
 D) $\theta < 45^\circ$ and a frictional force acts on the block towards Q.

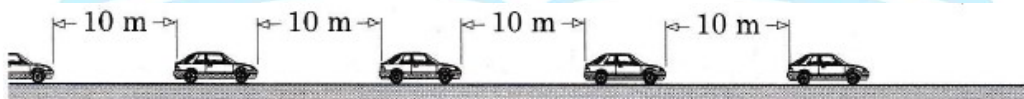
SECTION-III (Paragraph Type)

This section contains 2 groups of question. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choice (A), (B), (C) AND (D) for its answer, out of which ONLY ONE is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

Paragraph for Questions 33 and 34:

In a convoy on a long straight level road, 50 identical cars are at rest in a queue at equal separation 10m from each other as shown.

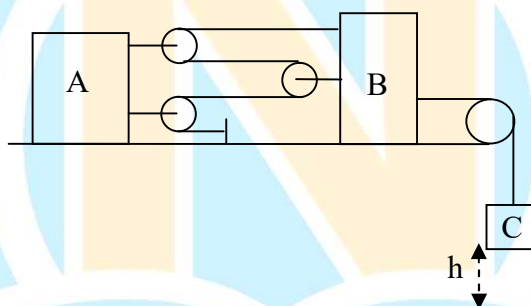


Engine of a car provide a constant acceleration 2 m/s^2 and brakes can provide a maximum deceleration 4 m/s^2 . When an order is given to start the convoy, the first car starts immediately and each subsequent car start when its distance from a car that is immediately ahead becomes 35m. Maximum speed limit on this road is 72 km/h. When an order is given to stop the convoy, the driver of the first car applies brakes immediately and driver of each subsequent car applies brakes with a certain time delay after noticing brake light of the front car turned red.

33. When all the cars are moving at the maximum speed, what is the separation between two adjacent cars?
- A) 35 m B) 85 m C) 100 m D) 110 m
34. During the time when motion is building up in the convoy, some of the cars are moving and the others are at rest. What is the average rate of change in length of the segment consisting of stationary cars?
- A) Decreasing at 0.5 m/s B) Decreasing at 1 m/s
C) Decreasing at 2 m/s D) Decreasing at 5 m/s

Paragraph for Questions 35 and 36:

Three blocks A, B and C are connected as shown in figure. All the surfaces are smooth. All the string and pulley are ideal. Mass of each block is m . Block C moves a distance h in vertically downward direction.



35. Velocity of the block A is
- A) $4\sqrt{\frac{2gh}{41}}$ B) $\sqrt{\frac{4gh}{41}}$ C) $\sqrt{\frac{18gh}{41}}$ D) $\sqrt{\frac{17gh}{41}}$
36. $\vec{v}_{AB}$ is the velocity of A w.r.t. B and $\vec{v}_{BC}$ is the velocity of B w.r.t. C, then
- A) $|\vec{v}_{AB}| > |\vec{v}_{BC}|$ B) $|\vec{v}_{AB}| = |\vec{v}_{BC}|$ C) $|\vec{v}_{AB}| < |\vec{v}_{BC}|$ D) $|\vec{v}_{AB}| = 0$

SECTION-I

(Single Correct Answer Type)

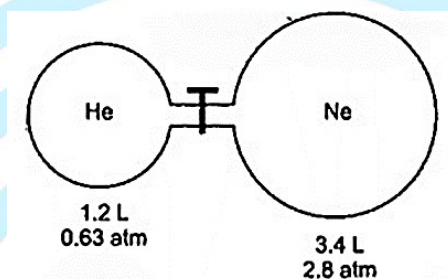
This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

37. 20 mL of 0.1 M solution of compound $Na_2CO_3 \cdot NaHCO_3$ is titrated against 0.05 M HCl, x mL of HCl is used when phenolphthalein is used as an indicator and y mL of HCl is used when methyl orange is the indicator in two separate titrations. Hence (y-x) is:

A) 40 mL B) 80 mL C) 120 mL D) None of these

38. Consider the following apparatus. Calculate the partial pressure of helium after the opening valve. The temperature remains constant at 16°C .



A) 0.164 atm B) 1.64 atm C) 0.328 atm D) 1 atm

39. Which of the following statement is incorrect?

A) In an isothermal process, temperature of a system remains constant but internal energy need not be
 B) C_p is almost equal to C_v for solids in general
 C) C_p is always greater than C_v for all types of gases
 D) In a cyclic process work done is zero but internal energy change is not equal to zero

40. Which of the following is not a conjugate acid-base pair?

A) HCO_3^- , CO_3^{2-} B) $C_6H_5^+NH_3$, $C_6H_5NH_2$
 C) $H_2C_2O_4$, $HC_2O_4^-$ D) OH^- , H^+

41. For the reaction, $A_{(g)} + B_{(g)} \rightleftharpoons C_{(g)} + D_{(g)}$, the initial concentration of A and B are equal, but the equilibrium concentration of C is twice that of equilibrium concentration of A.

The equilibrium constant is:

A) 8 B) 4 C) 2 D) 0.5

42. How many of the following are “intensive Properties”?

Pressure, Specific Heat, Molar Enthalpy, Molar Concentration, Hardness of Matter, Specific Volume, Specific Gibb’s Free Energy, Boiling Point, Density.

- A) 6 B) 7 C) 8 D) 9

SECTION-II

(Multiple Correct Answer Type)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases.

43. Which of the following statement is/are correct

- A) Change in entropy of a system in an irreversible adiabatic compression of ideal gas is positive
- B) During a reversible isobaric expansion of ideal gas temperature of system changes
- C) If two containers each filled with one mole of Hydrogen gas and Helium gas are mixed together at constant temperature results in increase of entropy of system
- D) conversion of solid white phosphorous to solid red phosphorous at its transition temperature is an example of phase transformation

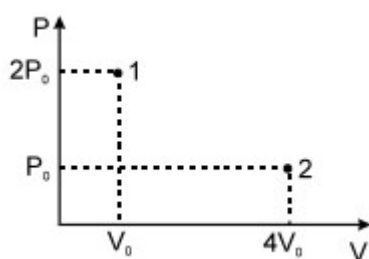
44. At constant temperature in a closed vessel of rigid volume addition of gaseous reactant/product will be completely nullified in which of the following equilibrium.

- A) $\text{NH}_4\text{HS}_{(s)} \rightleftharpoons \text{NH}_{3(g)} + \text{H}_2\text{S}_{(g)}$
- B) $\text{H}_2\text{O}_{(l)} \rightleftharpoons \text{H}_2\text{O}_{(g)}$
- C) $\text{LiCl} \cdot 3\text{NH}_{3(s)} \rightleftharpoons \text{LiCl} \cdot 2\text{NH}_{3(s)} + \text{NH}_{3(g)}$
- D) $\text{C}_{(s)} + \text{CO}_{2(g)} \rightleftharpoons 2\text{CO}_{(g)}$

45. Which of the following mixtures in aqueous medium on neutralization will show a same rise in temperature?

- A) 60mL of 0.1 M HCl+60 mL of 0.1 M NaOH
- B) 30mL of 0.1 M HCl+30 mL of 0.1 M NaOH
- C) 40mL of 0.1 M HCl+30 mL of 0.1 M NaOH
- D) 60mL of 0.1 M HCl+100 mL of 0.1 M NaOH

46. At TK addition of catalyst into the reversible reaction
- does not change the kinetic energy of reacting species
 - does not change the equilibrium constant of reaction
 - convert non spontaneous reaction into spontaneous reaction
 - does not change the enthalpy of reaction
47. The thermal dissociation equilibrium of $\text{CaCO}_3(s)$ is studied under different conditions
- $$\text{CaCO}_3(s) \rightleftharpoons \text{CaO}(s) + \text{CO}_2(g)$$
- For this equilibrium, the correct statement(s) is (are)
- ΔH is dependent on T.
 - K is independent of the initial amount of CaCO_3
 - K is dependent on the pressure of CO_2 at a given T
 - ΔG^θ is dependent on T
48. Select the correct statements :
- The entropy of an isolated system can never decrease.
 - For a chemical system in thermodynamic equilibrium at standard conditions, standard Gibbs energy change (ΔG^θ) T, P is equal to zero.
 - If the isentropic process for an ideal gas is reversible, it must be adiabatic.
 - A process occurs in such a direction as to cause an increase in the entropy of the universe.
49. A liquid which is confined inside an adiabatic piston is suddenly taken from state 1 to state 2 by a single stage process. If the piston comes to rest at point 2 as shown. Then



identify correct option(s)

- $W = -3P_0V_0$
- $\Delta U = -3P_0V_0$
- $\Delta H = -P_0V_0$
- $q = 0$

50. One way of writing the equation of state $PV = RT \left[A + \frac{B}{V} + \dots \right]$ Which is true for constant A & B in equation, if van der Waal's equation of a state is expressed in the above form
- A) expression for B is $a - \frac{b}{RT}$
- B) expression for B is $b - \frac{a}{RT}$
- C) unit of B is cm^3 and it is temperature dependent
- D) constant A must be 1

SECTION-III (Paragraph Type)

This section contains 2 groups of question. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choice (A), (B), (C) AND (D) for its answer, out of which ONLY ONE is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

Paragraph for Questions 51 and 52:

Oleum is considered as a solution of SO_3 in H_2SO_4 , which is obtained by passing SO_3 in solution of H_2SO_4 . When 100 g sample of oleum is diluted with desired weight of H_2O then the total mass of H_2SO_4 obtained after dilution is known as % labelling in oleum.

For example, a oleum bottle labeled as '109% H_2SO_4 ' means the 109g total mass of pure H_2SO_4 will be formed when 100g of oleum is diluted by 9 g of H_2O which combines with all the free SO_3 present in oleum to form H_2SO_4 as $\text{SO}_3 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4$.

Question:

51. If excess water is added into a bottle sample labelled as "112% H_2SO_4 " and is reacted with 5.3 g Na_2CO_3 , then find the volume of CO_2 evolved at 1 atm pressure and 300 K temperature after the completion of the reaction:
- A) 2.46 L B) 24.6L C) 1.23 L D) 12.3 L
52. 1 g of oleum sample is diluted with water. The solution required 54 mL of 0.4 N NaOH for complete neutralization. The % of free SO_3 in the sample is (nearly):
- A) 74 B) 26 C) 20 D) None of these

Paragraph for Questions 53 and 54:

A vessel of 25 lit contains 20 gm ideal gas 'X' at 300 K. The pressure exerted by the gas is 1 atm. 20 gm of ideal gas 'Y' is added to the vessel keeping same temperature. Total pressure became 3 atm. Upon further addition of 20gm of ideal gas 'Z' the pressure became 7 atm. Then answer the following questions

53. The molar mass of gas 'Y' is nearly

- A) 10 gm B) 20 gm C) 15 gm D) 5 gm

54. Identify the correct statement

I. Gas 'Y' is lighter than 'X'

II. Gas 'Z' is lighter than 'Y'

III. Gas 'X' is heavier than 'Z'

A) Only I and II are correct

B) Only I and III are correct

C) Only II and III are correct

D) All are correct

